

ORIGINAL RESEARCH

Modified case-crossover estimators were unbiased in the presence of temporal trends in the exposure of interest: a simulation study

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Abstract

Objectives: Temporal trends in exposure are a common source of bias in the case-crossover design. We evaluated four different estimators of the exposure odds ratio (OR) that adjust for such trends.

Methods: In a simulation study, we simulated 1 million individuals in a 5-year observation period and performed daily Bernoulli-trials to determine exposure status. Exposure probabilities followed a constant, increasing or decreasing trend, or one of three empirical trends based on real-world prescription drug use. The outcome rate was modified by a rate ratio (RR) of 0.5, 1.0 or 2.0. We tested four case-crossover estimators: a naive, unadjusted estimator; two “flipped” estimators modeling the probability of exposure instead of the outcome, adjusting for time linearly or using splines; and a new estimator (self-case-time-control [CTC]), adjusting the naive OR using an estimated trend 1 year prior. For reference, we also applied the CTC design. We performed 2500 simulations for each combination and obtained the mean OR, bias (measured as $\log(RR) - \log(OR)$), root mean squared error (RMSE), and coverage. Lastly, we applied all estimators to an empirical example assessing the association between semaglutide and retinal detachment (assumed null-effect) in Danish nationwide health registries.

Results: The naive case-crossover estimator was biased under in- or decreasing trends (bias -0.60 to $+1.18$). The two flipped estimators were unbiased under increasing trends (bias -0.09 to $+0.14$) but the spline estimator performed better under decreasing trends (bias 0.00 to -0.06). The self-CTC estimator performed similar to the linear estimator. The CTC design was unbiased in all scenarios and yielded the lowest bias. The empirical example yielded biased ORs for the naive estimator, while all other estimators yielded point estimates and 95% confidence intervals (CIs) that were compatible with a null effect.

Conclusion: The flipped and self-CTC estimators can be used to adjust for temporal trends in the case-crossover design. While the CTC estimator performed better, it requires the use of external time-controls and the assumption that the exposure trend is similar among cases and time controls. © 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Self-controlled designs; Observational studies; Simulation; Pharmacoepidemiology; Bias; Case-crossover design

1. Introduction

The case-crossover design is a self-controlled study design that compares the probability of being exposed at the time of an event of interest to the probability of being exposed at a reference time point. This design conditions the analysis on the individual, thereby eliminating confounders that are stable over time [1].

A core assumption of the case-crossover design is that the population's probability of exposure is stable across the observation period. This assumption often fails when the exposure of interest is a prescription drug, as new drugs enter the market and old drugs are replaced. Drug treatment is commonly assessed using prescription fills in windows of multiple months sometimes spanning a year between exposure assessment windows. These prolonged observation periods increase susceptibility to time trends.

The most common method used to address temporal trends in the exposure of interest in the case-crossover design is the case-time-control (CTC) design [2], an extension of the case-crossover design which makes use of a matched control group of noncases that is presumably

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What is new?

Key findings

- Temporal trends in exposure can result in substantial bias in the case-crossover design, and we evaluated the performance of one naive and three estimators adjusting for temporal trends and included the self-time-control estimator for comparison.
- In our simulation study, the naive estimator was biased among all scenarios with increasing or decreasing trends, whereas the flipped-linear, flipped-spline, and self-case-time-control (CTC) estimators substantially reduced this bias.
- The CTC estimator performed better than estimators in the case-crossover design in the simulation setting; however, its assumptions may be violated in a real-world setting.

What this adds to what is known?

- We provided the first formal evaluation of different estimators adjusting for temporal trends in exposure in the case-crossover design and compared to the CTC design.
- Specifically, we evaluated a newly proposed estimator coined the self-CTC estimator adjusting the naive OR using an estimated trend 1 year prior.

What is the implication/what should change now?

- Researchers should routinely adjust for temporal trends in exposure in the case-crossover design.

susceptible to the same time-related biases as the cases. Previous work has indicated that this assumption is not always met [3], and further extensions of the CTC design have been developed to increase comparability of cases and controls, such as the case-case-time-control design [4], where future cases serve as the matched control group.

An alternative to using a control group is to define multiple reference windows and modeling a time variable, as described by Suissa [2] and Allison and Christakis [5]. This method requires the specification of two or more reference windows and application of the original specification of the CTC model, where exposure status is used as the dependent variable and outcome status and time as explanatory variables. Suissa introduced two different parametrizations of the CTC model in the original paper, the above-described exposure model and an outcome model based on conditional logistic regression. To avoid confusion, we will refer to the approach based on an exposure model as the flipped

case-crossover estimator and refer to the outcome model as the CTC estimator.

The flipped case-crossover estimator has not been formally evaluated in the context of case-crossover studies in pharmacoepidemiology. Therefore, we aimed to investigate the performance of the naive case-crossover, the flipped case-crossover adjusted for calendar time, and a novel estimator coined the self-CTC under different temporal trends of the exposure of interest. For reference, we also applied a CTC design, including the sampling of time-controls.

2. Methods

In this simulation study, we evaluated bias and precision of four different estimators when analyzing a case-crossover study and the CTC design in the presence of temporal trends in the exposure of interest. We simulated six different scenarios with varying temporal trends in exposure. For comparison, we also applied the CTC design. We performed 2500 simulations for each combination of data-generating mechanism and estimator and obtained the mean odds ratio (OR), bias, root mean squared error (RMSE), and coverage with corresponding Monte Carlo standard errors (MCSE). Lastly, we applied all estimators to an empirical data example and assessed the association between use of semaglutide and retinal detachment in the Danish nationwide health registries.

2.1. Data generating mechanism

We simulated a population of 1,000,000 initially unexposed individuals who were under observation for a period of 5 years (1825 days). We then simulated an episodic exposure of interest, reminiscent of prescription drugs used for short-term treatments, such as antibiotics for bacterial infections or corticosteroids for chronic obstructive pulmonary disease. For each individual, on each given day, we performed a Bernoulli trial to determine whether the individual had a prescription fill on that day. This allowed us to freely specify the rate of exposure for a given day and, henceforth, the shape of the temporal trend. All individuals were exposed from the first day of a given prescription fill and for the following 120 days. It was possible to have overlapping prescription fills, resulting in extended exposure.

For the scenario with no temporal trend, all days of the study period were assigned the same probability of prescription (1/73,000, corresponding to 1 treatment episode per 200 person years). For the increasing trend, we generated probabilities of exposure based on a half-normal distribution, with the mode of the distribution in 1825 and a standard deviation of 1825/4, with the rate of exposure on day 1 of the study period being 1 per 200 person years and increasing five-fold to a daily rate of exposure of

1/40 person years on day 1825. Correspondingly, for the scenario with the decreasing trends, the rate of exposure decreased from 1 per 200 person years on day 1 to 1 per 1000 person years on day 1825. For a graphical depiction of simulated trends, see Figure 1A–C.

Further, we simulated exposure times based on observed probabilities of exposure obtained from the Danish National Prescription Registry [6] for three different types of drugs: semaglutide (rapid drug uptake), sertraline (nonmonotone relationship with both increasing drug uptake and a decline), and warfarin (decreasing prevalence of use). For a graphical depiction of these trends, see Figure 1D–F.

Finally, we simulated event times for a binary outcome with a rate of one event per 100 person years. One outcome time was drawn for each individual from an exponential distribution. The rate of the outcome was modified by a rate ratio (RR) for the exposure of interest, which could be 0.5, 1.0, or 2.0. The outcome rate was modified by the RR during the 120 days following the prescription fill. As sensitivity analyses, we considered a duration of exposure of 60 days following a prescription fill, varied the baseline rate to 1 treatment episode per 100 and 400 years and shortened the lookback period for control windows, such that the reference (nonfocal) windows ended on day 180 and 360 prior to the event.

2.2. Estimand

The estimand of interest in this simulation study is the true RR that modifies the outcome rate in the generative model described in the above section.

2.3. Analysis

Data were analyzed using a case-crossover design as the primary design of interest, and the CTC design was included for reference. In brief, after we simulated exposure and outcome times, we restricted the study sample to all individuals who had an outcome before the end of the observation period. We then specified three exposure assessment windows with a duration of 120 days, to assess the exposure status at the end of each window. One (the focal window) ending on the day prior to the event of interest, and two reference (nonfocal) windows ending one and 2 years (365 and 730 days) prior to the event of interest. We considered an individual exposed if a prescription fell within one of the exposure assessment windows. To allow for sufficient lookback for all windows, we only considered individuals where the event occurred on day 850 or later. For the CTC design, we additionally sampled five controls per case from individuals who had not experienced the event during the observation period. Controls were assigned an index date identical to that of their corresponding case, and had discordant exposure status for their index and reference dates. Using the above data structure, we evaluated five different estimators:

- 1) The naive case-crossover estimator: Conditional logistic regression with type of window (focal/nonfocal) as the dependent variable and exposure status as an explanatory variable, conditional on the individual. Only the focal window and first reference window were used to estimate the exposure OR, according to the original paper by Maclure [1].

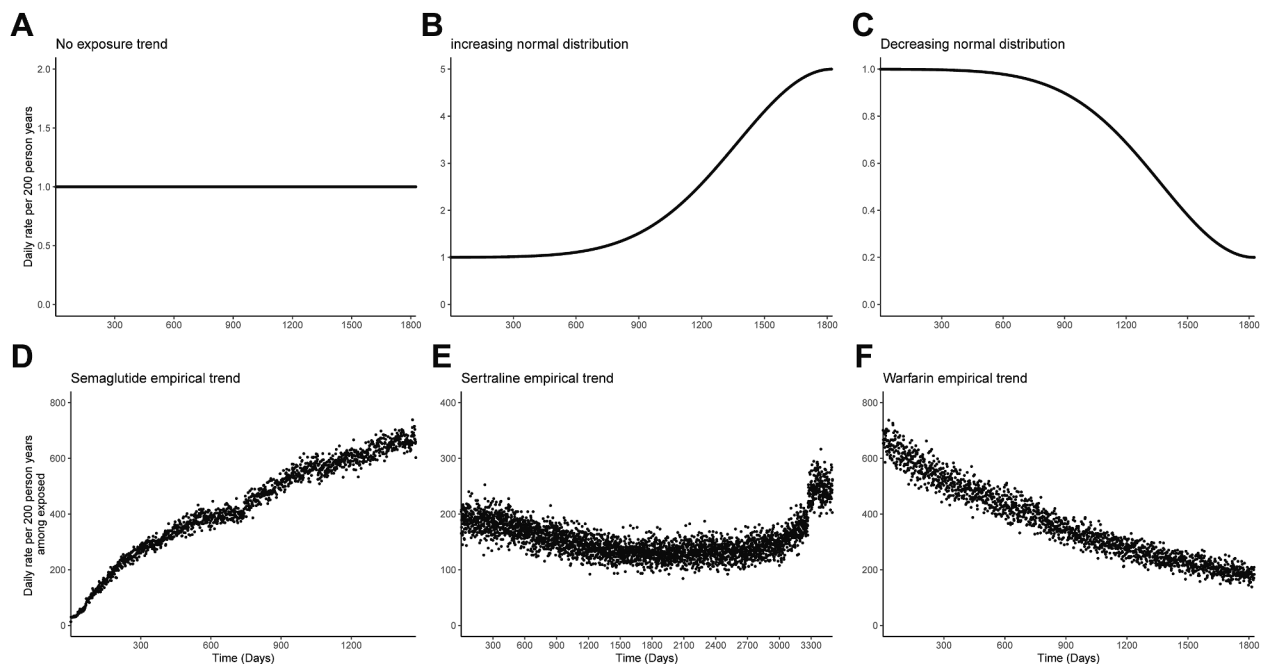


Figure 1. Simulated exposure time trends (A–C) and empirical exposure time trends for selected medications (D–F).

- 2) The flipped case-crossover estimator with time included as a linear covariate: Conditional logistic regression with exposure status as the dependent variable and type of window and calendar time as explanatory variables. The regression was conditioned on the individual and made use of all three time windows (focal window and reference window 1 and 2).
- 3) The flipped case-crossover estimator but with calendar time modeled by restricted cubic splines with four knots placed at the fifth, 35th, 65th, and 95th percentile of the observed start times of each exposure assessment window [7].
- 4) The self-CTC: The exposure OR estimated using the naive case-crossover estimator is divided by an exposure OR representing the temporal trend for the time between the first and the second reference (nonfocal) window. The trend exposure OR is estimated from a dataset containing only individuals with a discordant exposure status in reference window 1 and 2. The trend OR is calculated as the number of individuals exposed in reference window 1 divided by the number of individuals exposed in reference window 2. In essence, this corresponds to a CTC design, where each subject with an outcome contributes one case set (focal and referent window) and one control set (focal and referent window) 1 year prior to the outcome. Unlike the classical CTC, the pool of case sets and control sets are analyzed separately in the self-CTC. Estimation of the variance for the self-CTC estimator was achieved using a modified Woolf logit [8] (Appendix 1).
- 5) The CTC estimator: Conditional logistic regression with type of window (focal/nonfocal) as the dependent variable, and exposure status, case status, and their interaction as explanatory variables, conditioning on the individual. The interaction terms allow for estimating the OR adjusted for temporal trends.

2.4. Performance measures

For each scenario, we ran 2500 simulations, and from these we calculated the mean estimate, bias, RMSE and the coverage of the 95% confidence intervals (CIs) for all described estimators. Further, we calculated MCSE for all performance metrics [9]. The source code and empirical data used to perform this simulation study are available at <https://gitlab.sdu.dk/segsgaard/trend-cco>.

2.5. Empirical data example

We also applied all estimators to data obtained from the Danish nationwide health registries, to evaluate their performance when faced with the complexities of real-world data. We estimated the exposure OR for filling a semaglutide prescription in relation to receiving a diagnosis of

retinal detachment. We chose semaglutide (ATC [Anatomical Therapeutic Chemical Classification] code A10BJ06) as the exposure of interest, as it is a drug with a recent surge in the number of users. Retinal detachment was chosen as the outcome of interest, as it is not known to be related to semaglutide and therefore should yield null results.

We identified all incident diagnoses of retinal detachment (ICD-10 [International Classification of Diseases 10th revision] code H33) from the Danish National Patient Register and considered these individuals cases. We excluded cases with a recorded diagnosis that was within the first 850 days after the first recorded prescription date of semaglutide. Focal and reference windows were defined as outlined in the simulation study and individuals were considered exposed if a prescription fill was recorded in the Danish prescription registry during one of the exposure assessment windows. For the CTC design, we sampled up to four time-controls with a discordant exposure status. Time-controls were matched to cases according to calendar time, birth year, and sex.

For all estimators, we obtained the exposure OR and corresponding 95% CIs using conditional logistic regression.

According to Danish law, ethical approval is not needed for purely register based studies. The study was approved by the local data protection board (no. 11.570) and the Danish Health Data Authority (FSEID-00006047).

All programming was performed using R version 4.4.2 [10] and the data.table [11], doFuture [12], Epi [13], rms [14], and rsimsum [15] packages.

3. Results

For all scenarios, we obtained satisfactory MCSE with 2500 replications, with the MCSE for bias being below 0.01 in 67 out of 90 scenarios (74%) and a maximum MCSE of 0.018 (Supplementary Table S1). All estimators were unbiased in the scenario with a constant probability of exposure throughout the observation period (bias between -0.04 and 0.01) (Fig2).

When we simulated an increasing trend in the exposure probability (Fig 1B), the naive estimator was upwards biased across all simulated RRs (bias 0.55 – 0.56), while the other estimators were unbiased (-0.03 to 0.01) and with comparable RMSEs (0.11 – 0.12 for the linear and self-CTC estimator, 0.16 – 0.19 for the spline estimator, 0.03 – 0.05 for the CTC estimator). We observed substantial bias in the scenario with a decreasing trend (Fig 1C) for the naive estimator (-0.60 to -0.54) and the linear and self-CTC estimator (-0.36 to -0.38), but not for the spline estimator (-0.02 to -0.05) and the CTC estimator (-0.03 to -0.07). For all estimators, bias increased with a decreasing RR. RMSEs were comparable across all case-crossover estimators (range 0.39 – 0.85) but lower for the CTC estimator (0.13 – 0.34).

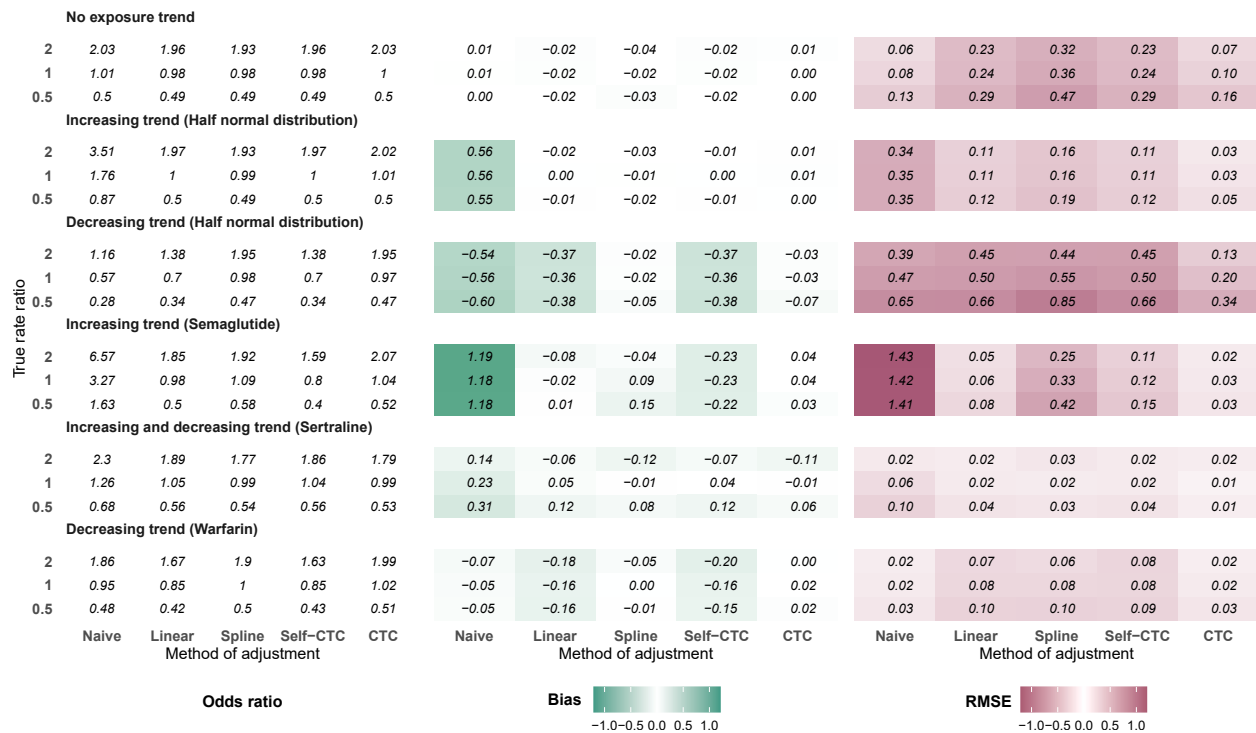


Figure 2. Odds ratio, bias, RMSE of naive, linear, spline, self-CTC and CTC exposure time trend adjustment of simulated and empirical scenarios under varying risk ratios. RMSE, root mean squared error.

For more complex empirical trends, we saw similar results to the trends based on the half-normal distribution. Results for the scenario based on the exposure probability of semaglutide (Fig 1D) were similar to the increasing trend based on the half-normal distribution, except that bias was amplified for the naive estimator (1.18 and 1.19) and the self-CTC estimator being moderately downwards biased (-0.22 and -0.23). The lowest RMSE was observed for the CTC-estimator (0.02–0.03) followed by the linear (0.05–0.08), self-CTC (0.11–0.15), spline (0.25–0.42) and naive estimator (1.41–1.43). In the scenario with a nonmonotone trend based on sertraline prescriptions (Fig 1E), all estimators exhibited slight bias (-0.12 to 0.12), except for the naive estimator (0.14–0.31), with no substantial differences in the RMSE. Finally, the decreasing trend based on warfarin prescriptions (Fig 1F) yielded moderate bias for the linear and self-CTC estimator (range -0.20 to -0.15) but did not affect the naive (-0.05 to -0.07), spline (-0.05 to 0.00), or CTC (0.00–0.02) estimator. For coverage, all estimators provided close to the nominal coverage (0.95) when unbiased (Supplementary Fig S1). The sensitivity analysis using an exposure window of 60 days following a prescription fill, varying baseline exposure rate and shortening the look-back period of control window, yielded similar results (Supplementary Figs S2–S5).

Findings from the empirical data example of semaglutide and retinal detachment yielded biased ORs for the naive estimator with an OR of 6.85 (95% CI, 5.20–9.03),

while all other estimators yielded point estimates and 95% CIs that were compatible with a null-effect (Fig 3). The CTC estimator with a standard error (SE) of the log OR of 0.19 was the most precise among the estimators compatible with a null effect. The self-CTC estimator was the least precise with an SE of 0.31.

4. Discussion

In this study, we evaluated the performance of four different estimators under the case-crossover paradigm in the presence of temporal trends in the exposure of interest and compared these to estimates obtained from a CTC design. All estimators were unbiased when the population probability of exposure was constant over time. We found that the naïve case-crossover estimator was biased across most scenarios with exposure trend, whereas the other estimators were unbiased or exhibited little bias. Among the estimators within the case-crossover design, the flipped case-crossover estimator with calendar time modeled as restricted cubic splines yielded the lowest bias across all scenarios but was less precise compared to the other estimators that accounted for calendar time using a linearity assumption. The CTC estimator consequently showed smaller bias than the estimators in the case-crossover design. In the empirical data example, the naive estimator yielded a substantially elevated OR, whereas all other estimates were compatible with the assumed null effect.

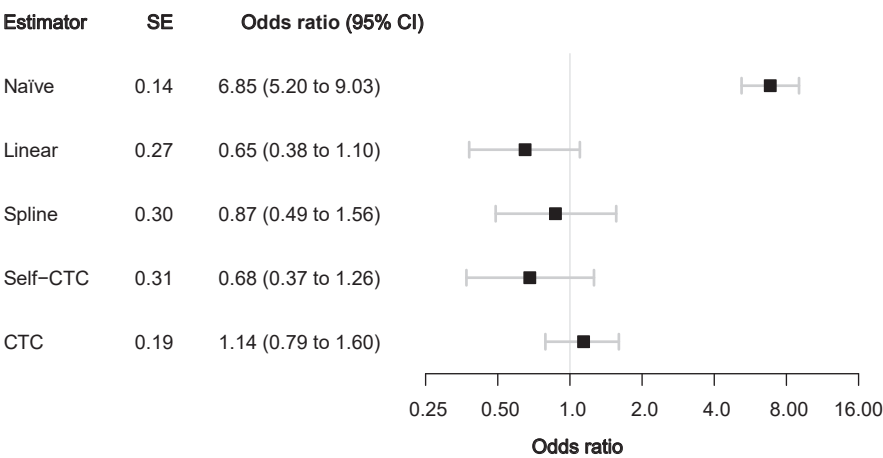


Figure 3. Empirical example on the association between semaglutide and retinal detachment using all five estimators. OR, odds ratio; CI, confidence interval; CTC, case-time-control.

The strengths of our simulation study are that we evaluated a range of estimators under multiple data generating mechanisms, with focus lying on the shape of the temporal trends instead of underlying probabilities of exposure, true RRs, or design specifications. Further, we made use of empirical distributions for the temporal trends in several scenarios to increase the relevance of our findings when applied to pharmacoepidemiological studies. While estimators within the case-crossover design were our primary design of interest, we further included estimates from a CTC design for reference. In addition, we included a range of sensitivity analyses varying the exposure window, baseline rates, and reference windows, which showed findings consistent with the main analysis, supporting the robustness and strengthening the applicability of the evaluated estimators. Albeit intentional, our study is limited to studying temporal trends in exposure but did not include other biasing factors such as time-varying confounding or differing temporal trends among cases and noncases. While including scenarios with time-varying confounding and how to adjust for this is an important and relevant topic, it falls out of the scope of the current study.

In this simulation study, the CTC estimators consistently outperformed the case-crossover estimators. This was expected given our data generating mechanism, where the cases and controls were subject to the same time-varying probability of exposure. Moreover, the CTC design gains higher precision by sampling up to five time-controls for each case, allowing more precise estimates. However, we argue that the underlying assumption, namely that the non-cases are susceptible to the same time-related biases as the cases, not necessarily holds in a real-world setting [16]. Thus, while the CTC design outperforms the case-crossover estimators in a simulation setting, this advantage may not necessarily translate to applied studies where such assumptions may be violated. Another estimator that can adjust for temporal trends in a case-crossover design,

besides those that we evaluated in this simulation study, is the generalized estimation method for the CTC design described by Sjölander and Ning [17]. The estimator proposed by Sjölander and Ning yielded similar results to the flipped estimators in preliminary analyses (data not shown). The advantage of this estimator is that it can be used to evaluate nonbinary outcomes, something that may be useful but outside the scope of our simulation.

In this simulation study, we showed that by using an appropriate estimator of the case-crossover design, one can easily adjust for monotone, continuous covariates under a case-crossover design, such as calendar time. The implications for current practice within the field of pharmacoepidemiology are clear: Applied pharmacoepidemiologists should adjust for calendar time in case-crossover analyses, instead of assuming that the probability of exposure is stable across the observation period. None of the adjusted estimators proved superior to the others. Still, we recommend the flipped case-crossover estimator using restricted cubic splines for modeling calendar time when evaluating whether any temporal biases are present. If it seems likely that the exposure trend is linear, the analyst can then gain precision by using a simpler modeling approach as in the flipped estimator with time as a linear covariate. The self-CTC estimator may be preferable in a high-throughput setting, such as large-scale screening for drug safety signals, where careful evaluation of each outcome-exposure pair is not feasible.

5. Conclusion

We found that the flipped case-crossover estimators and the self-CTC estimator can be used to adjust for temporal trends in the exposure of interest when using a case-crossover design. Further, we highlighted the importance of accounting for temporal trends, as these can, and will,

lead to substantial bias when considering drugs that are entering or leaving the market. While the CTC design outperformed the case-crossover estimators in a simulation setting, we argue that the assumption behind the CTC estimator may be violated in a real-world setting, limiting its applicability in applied pharmacoepidemiology. Future studies should address the impact of other temporal factors such as time-varying confounding and evaluate analytical approaches to appropriately account for these.

CRediT authorship contribution statement

Sofie Egsgaard: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Jesper Hallas:** Writing – review & editing, Methodology, Conceptualization. **Lars Christian Lund:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors have no conflict of interest and no financial disclosures.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112291>.

Data availability

Link for data has been shared.

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